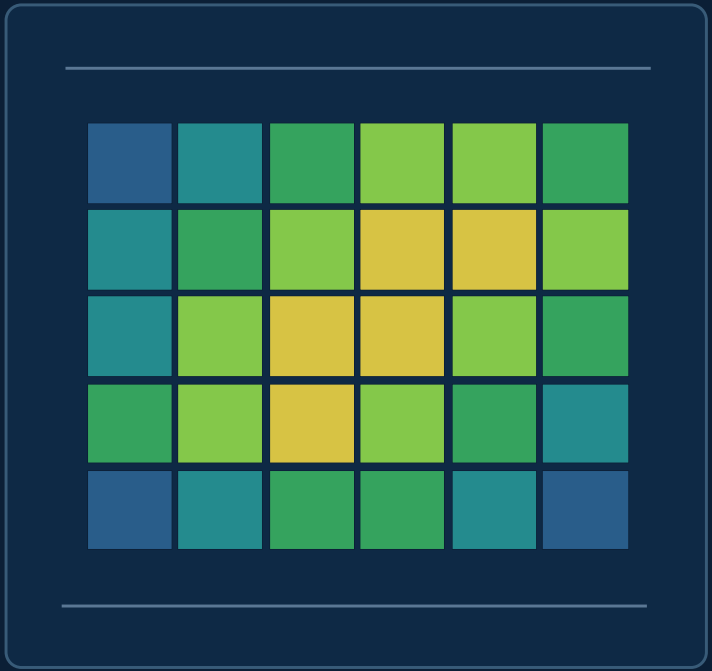
Crypto Alpha Across Horizons

Cross-Sectional Momentum, Reversal, and Trading Activity in a 24/7 Market

Owen Simon



The universe was point-in-time, liquid, and investable by construction

1 **Binance global spot USDT pairs** →
Direct exchange bars; no synthetic prices

2 **90 days of listing history** →
Eligibility filter before a coin can enter

3 **30d median daily quote volume** →
Liquidity ranking at each month-end

4 **Monthly top 25**
Point-in-time membership, not today's winners

76
usable months
Apr 2020–Jul 2026

25
assets per month
constant cross-section by design

151
unique research assets
union across monthly snapshots

Integrity checks covered schema, chronological order, duplicate/misaligned bars, nulls, point-in-time membership, and universe coverage before alpha research advanced.

The design guards against survivorship and liquidity hindsight by reconstructing membership month by month.

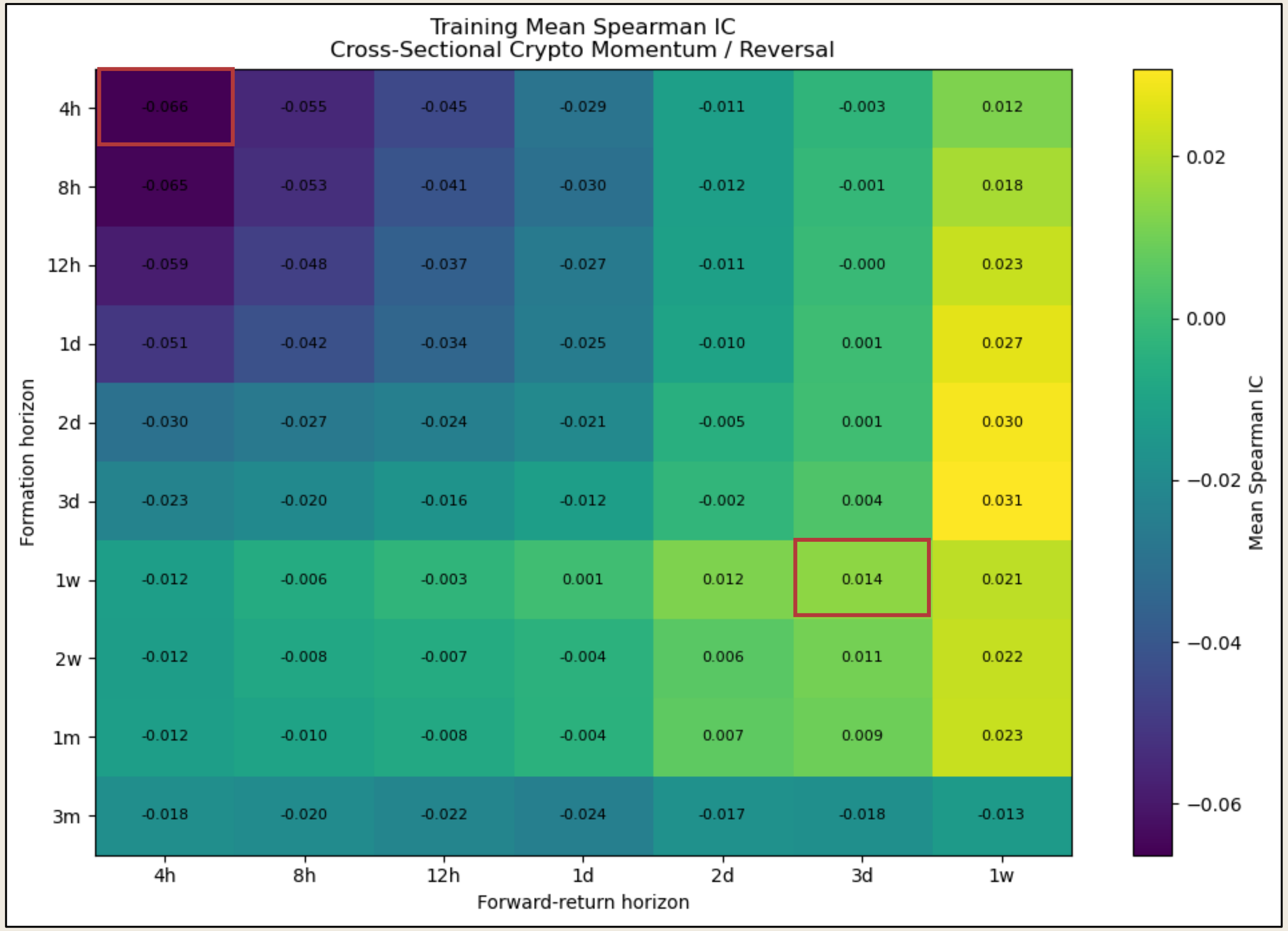
Predictiveness was only the first gate

Formation return	→	Cross-sectional IC	→	Neutral portfolio	→	Overlapping cohorts	→	Execution accounting	→	Validation gates
P(t) / P(t−F) − 1		Spearman(signal, future return)		gross-one long/short weights		holding horizon ≠ decision cadence		full-L1 turnover × cost		frozen rule before opening sample

Guardrail	Implementation in this project
No look-ahead	Weights earn subsequent returns only; 2025+ held sealed until final holdout validation
Transaction costs	20 bps per unit of full-L1 turnover is the primary model
Economic viability	Break-even cost used as an intuitive resilience metric
Overfitting control	Broad surfaces and frozen archetypes favored over one-cell optimization

The research sequence was built to separate predictive signal, implementable economics, and genuine out-of-sample evidence.

Initial research found two distinct effects across horizons



What the surface suggested

<u>Very-short reversal</u>

- negative IC at 4h–12h formation and short forward horizons
- statistically strong but turnover intensive

<u>Medium-horizon momentum</u>

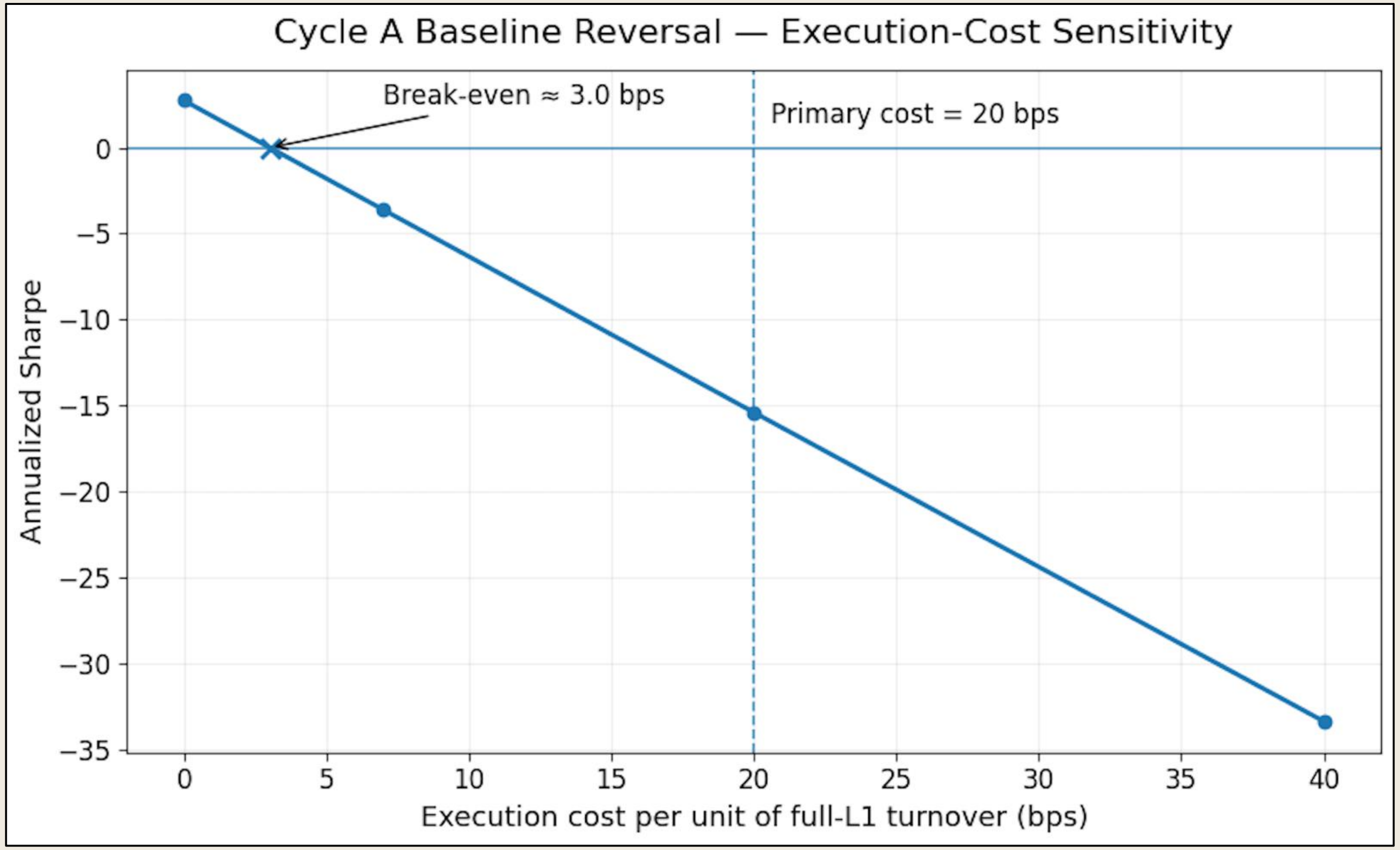
- positive cells around multi-day formation and 1w forward horizon
- advanced to portfolio validation

Activity conditioning was studied but not forced into the final architecture.

<u>4h formation → 4h forward</u> (short-horizon reversal) and <u>3d formation → 1w forward</u> (medium-horizon momentum) were chosen as the two core Cycle A candidates for deeper testing.

Strong predictiveness did not make reversal tradable

Reversal break-even costs stayed far below the 20-bps primary assumption



Baseline reversal looked strong before costs, with a gross Sharpe of 2.74, but its break-even execution cost was only ~3.0 bps. At the locked 20-bps assumption, Sharpe fell to −15.39.

Decision: reversal is research-complete and not promoted.

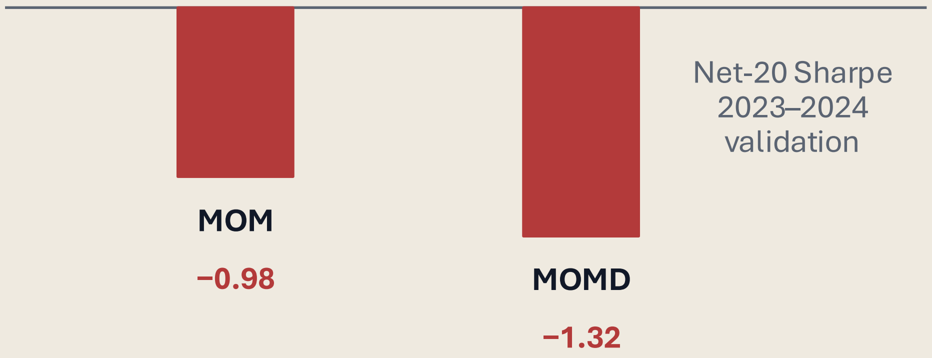
Reversal was statistically strong, but execution costs eliminated its economic viability.

The original momentum strategy failed its frozen 2023–2024 validation

VALIDATION FAILURE

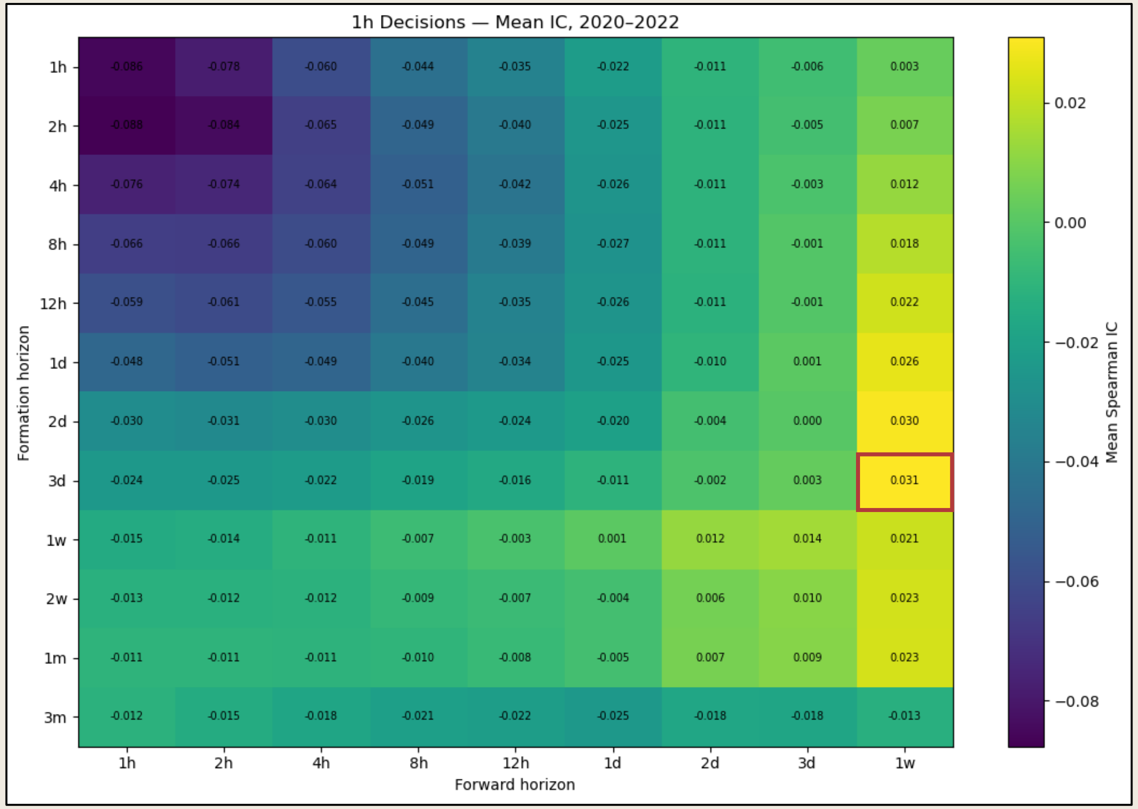
Neither MOM nor MOMD (MOM with 1-day execution delay) candidate passed the predeclared positive 20-bps-net-return viability rule. 2025+ remained closed.

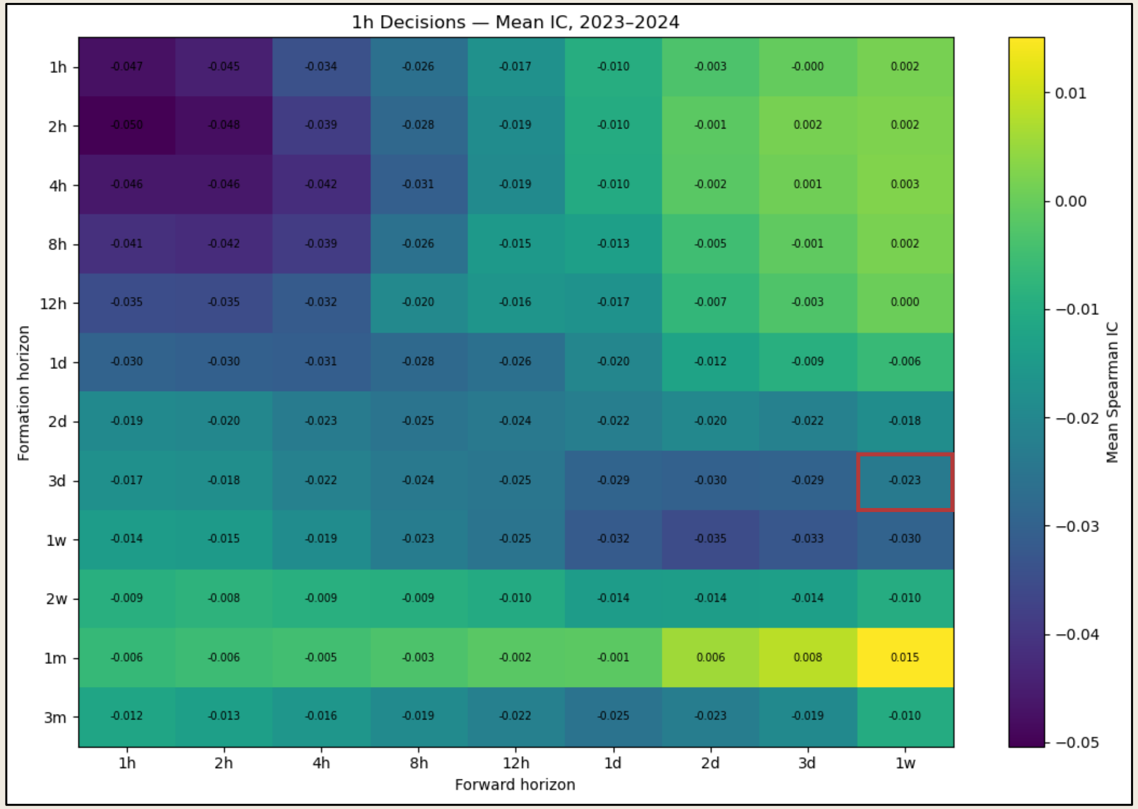
Candidate	Net-20 Sharpe	BE cost	Pass
MOM	−0.98	2.2 bps	False
MOMD	−1.32	−1.6 bps	False



The later continuation result is credible precisely because this failure was preserved, not overwritten.

The failure was structural: the original momentum relationship changed sign





2020–2022: positive 3d → 1w momentum cell

2023–2024: negative 3d → 1w momentum cell

The old 3d → 1w static momentum specification failed structurally: its predictive sign changed in validation.

The original research became Cycle A, as the failure prompted a separate, redesigned Cycle B

Cycle A: original 4h research cycle

VALIDATION FAILURE

2020–2022
Training discovery

2023–2024
Frozen validation

MOM/MOMD
Validation failed

2025+
Kept sealed

Cycle B: redesign using only pre-2025 evidence

Native 1h
Failure diagnosis

2020–2024
Continuation redesign

ECON/BAL/SHARP
Finalists frozen

2025–Jul 2026
Untouched holdout

The original failure remained part of the evidence; no 2025+ performance was used to rescue the signal.

The redesign separated signal horizon, holding horizon, and trading cadence

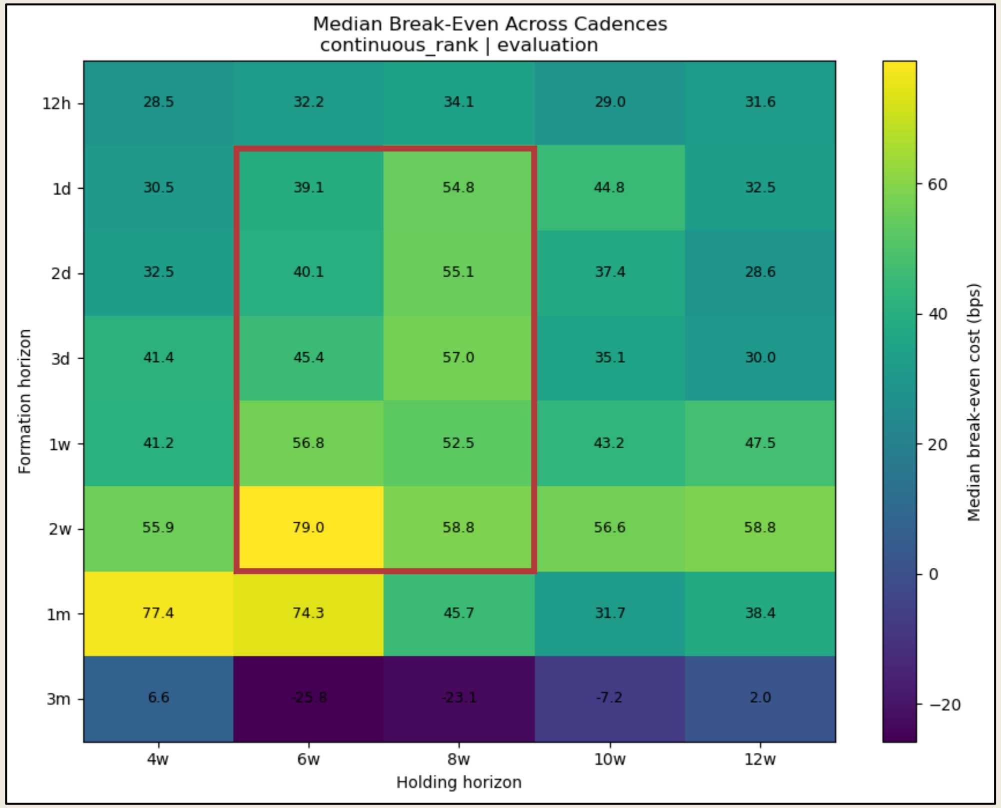
Native 1h source clock	→	Formation horizon	→	Cross-sectional rank / terciles	→	Decision cadence	→	Holding horizon	→	Execution economics
true exchange timestamp grid; missing hours retained as NaN	→	relative past performance signal	→	two mappings kept separate	→	fixed UTC refresh schedule	→	overlapping multi-week cohorts	→	full-L1 turnover and 20-bps primary cost

PRE-2025 DATA ONLY

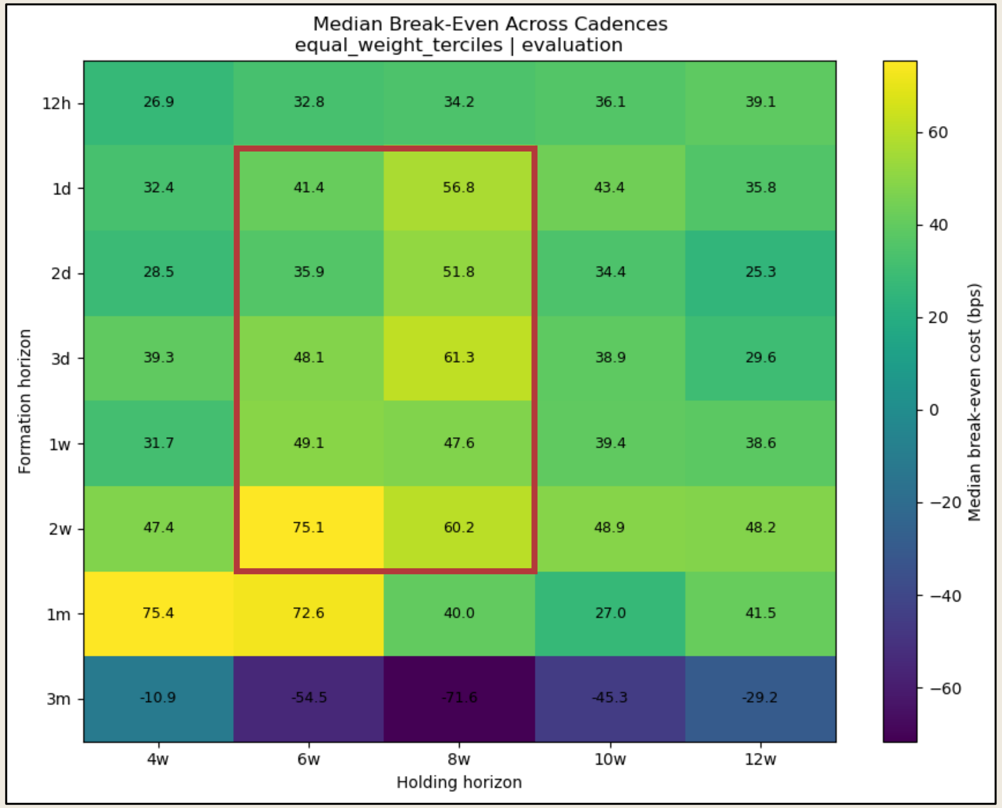
Key change: **holding horizon** and **trading cadence** became <u>implementation dimensions</u> rather than being conflated with the predictive signal horizon.

No reversal/continuation combination was tested after reversal failed economics; continuation was redesigned independently.

Continuation became economically viable in a broad interior region



Continuous-rank evaluation



Equal-weight-tercile evaluation

1m formation showed strong evaluation economics but was unstable across development/evaluation and was not promoted.

The viable region is not one optimized cell: roughly 1d–2w formation and 6w–8w central holding remain well above the 20-bps threshold, while 3m formation fails.

Three different implementation philosophies were frozen before 2025+ was opened

ECON	BAL	SHARP
Economic resilience	**Balanced**	**Higher Sharpe**
continuous rank	equal-weight terciles	equal-weight terciles
2w formation	3d formation	1d formation
6w hold	8w hold	8w hold
24h cadence	48h cadence	8h cadence
Eval BE 79.0 bps	**Eval BE 73.0 bps**	**Eval BE 60.9 bps**
Eval gross Sharpe 0.92	Eval gross Sharpe 1.42	Eval gross Sharpe 1.94

These are alternate implementations of one continuation factor, not hindsight-selected winners or three independent alphas.

The final test was opened once under frozen rules

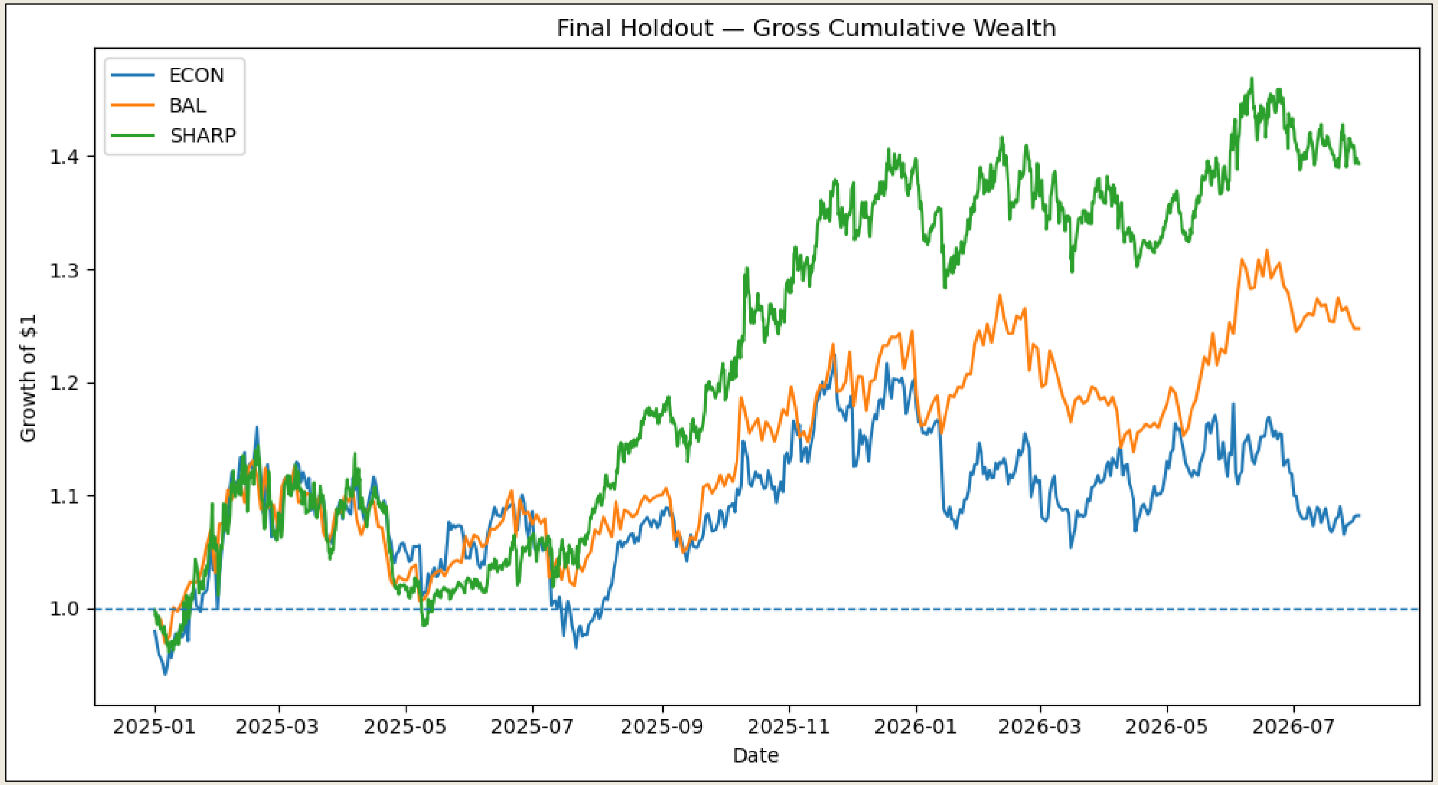
Redesign freeze	Reproduction gate	Open holdout once	Evaluate rule
ECON / BAL / SHARP definitions locked	pre-2025 metrics reproduced within tolerance	2025-01-01 through 2026-08-01 exclusive	IC > 0 + gross return > 0 + BE ≥ 20 bps

A post-holdout parameter rescue was not allowed

No combined portfolio, beta hedge, timing filter, asset exclusion, leg redesign, or winner-by-realized-Sharpe selection was added after seeing 2025+.

The final holdout was a confirmation test, not another search window.

All three frozen continuation implementations survived the untouched holdout



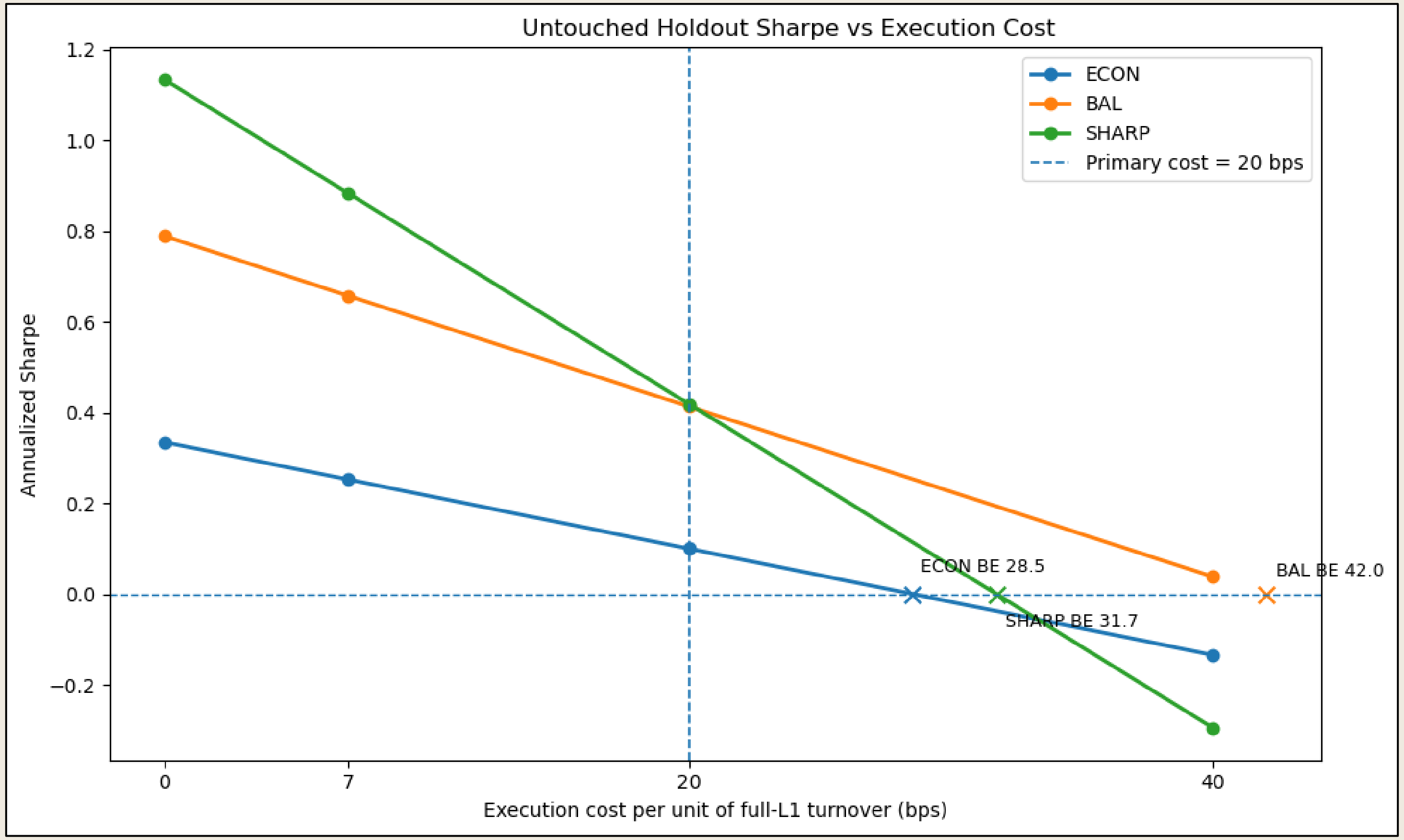
23.1%
SHARP gross ann. return
Gross Sharpe 1.13

42.0 bps
BAL break-even
Strongest cost headroom

3 / 3
primary survival pass
All frozen archetypes

Finalist	Mean IC	HAC t	Gross Sharpe	Net-20 Sharpe	BE cost
ECON	0.048	1.21	0.33	0.10	28.5 bps
BAL	0.047	2.27	0.79	0.41	42.0 bps
SHARP	0.037	2.59	1.13	0.42	31.7 bps

All three retained positive cost headroom above the locked 20-bps assumption

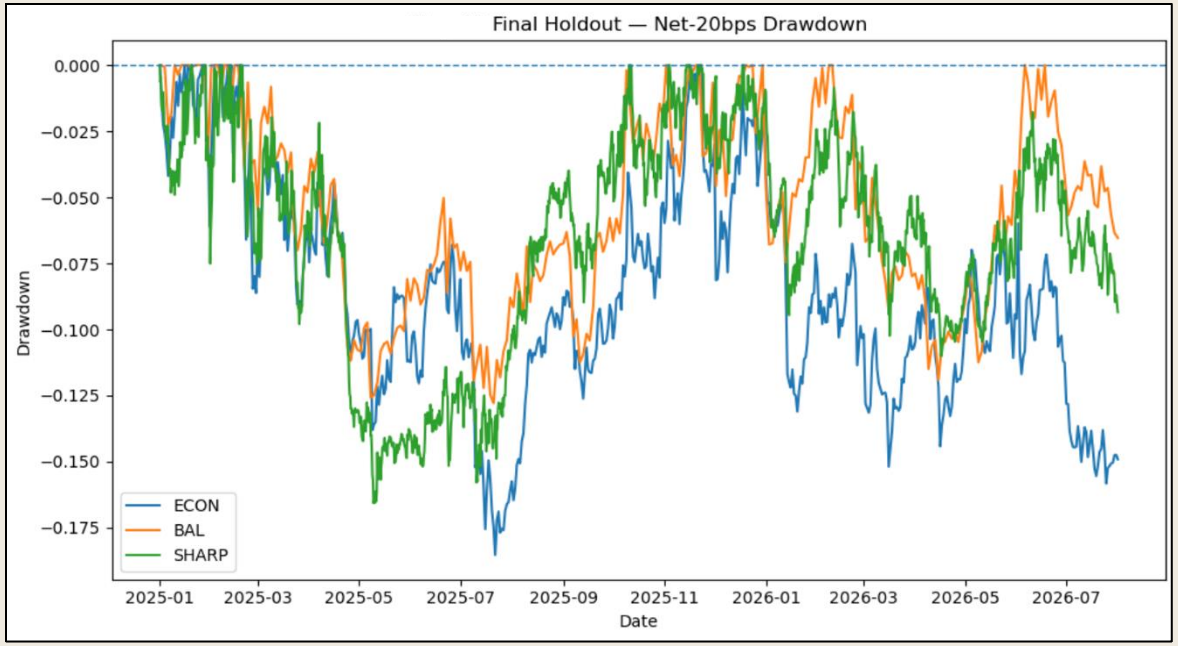


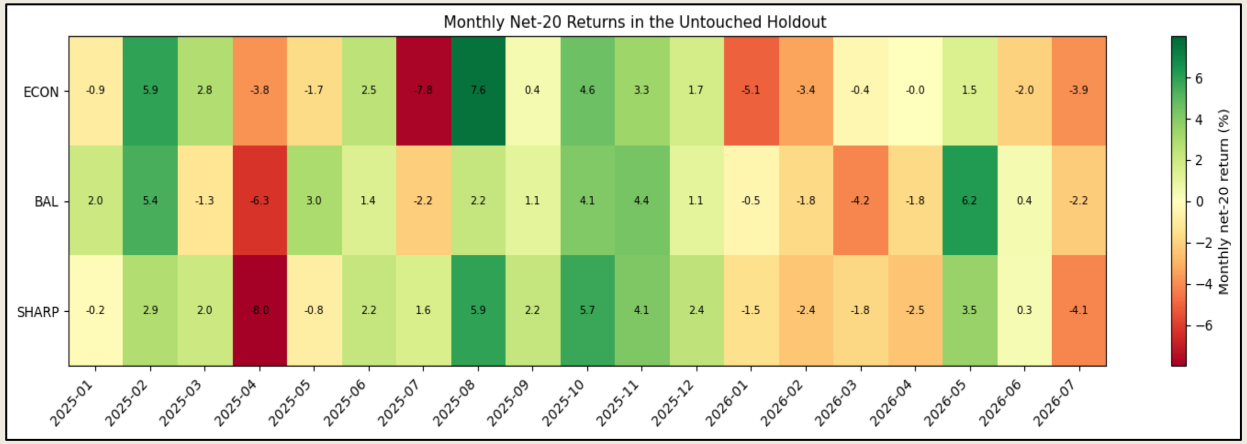
Holdout break-even headroom

ECON	+8.5 bps	26.3× turnover
BAL	+22.0 bps	38.2× turnover
SHARP	+11.7 bps	72.7× turnover

The margin compressed materially versus pre-2025 evidence; BAL was the strongest realized cost-resilience archetype.

Survival did not mean smooth or uniform performance



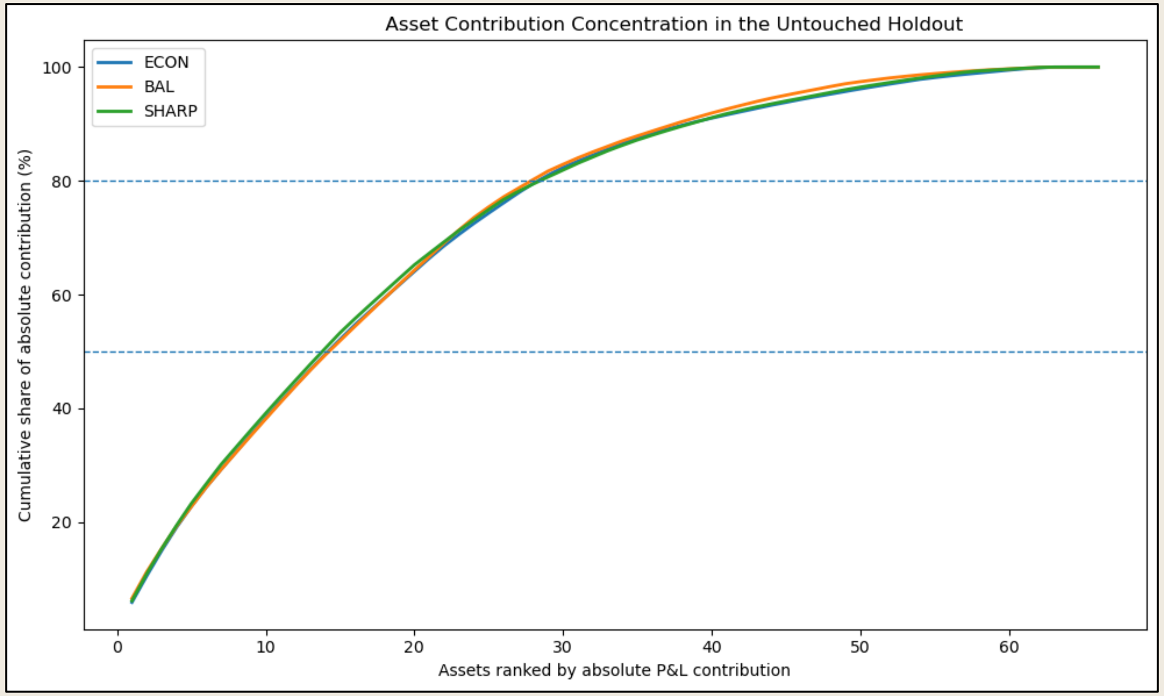


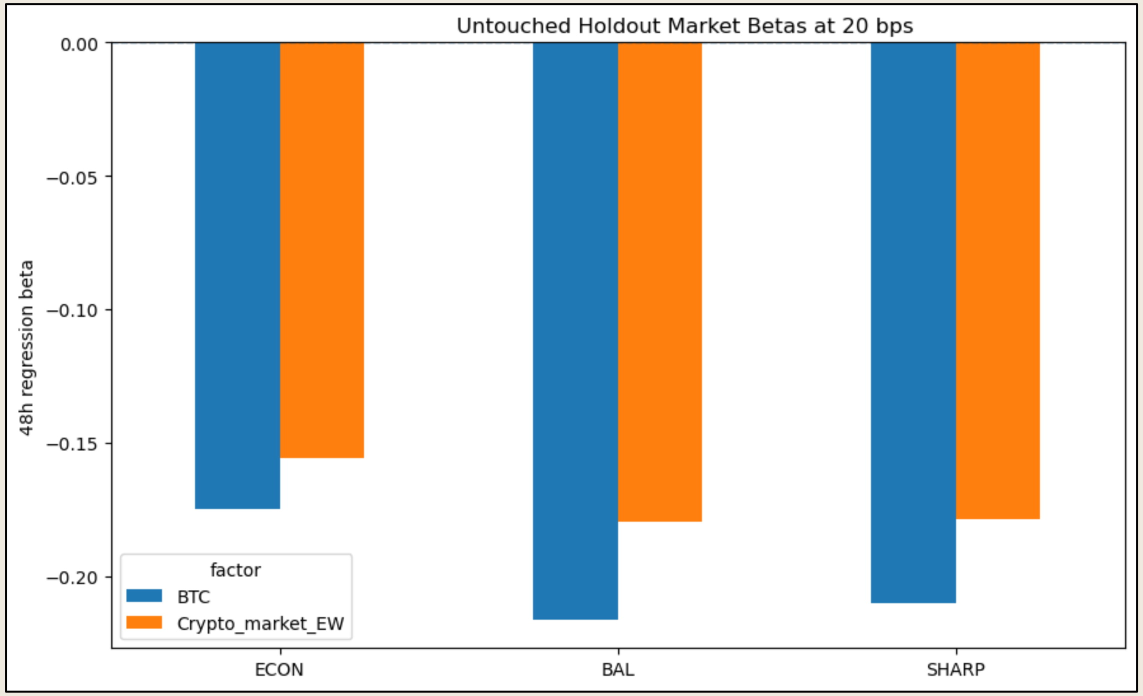
Finalist	Net-20 max DD	Worst 48h	Worst 8d
ECON	−18.5%	−6.6%	−8.1%
BAL	−12.8%	−4.9%	−6.6%
SHARP	−16.6%	−5.1%	−7.0%

Finalist	2025 net-20 Sharpe	2026 Jan–Jul net-20 Sharpe
ECON	0.69	−1.05
BAL	0.78	−0.28
SHARP	0.99	−0.75

The holdout result is positive in aggregate but marked by drawdowns, long underwater periods, and a weaker Jan–Jul 2026 segment.

The spread was broad, but not completely market-factor neutral





29-31
median active names

≤6.6%
largest single-name share

Despite near-zero net dollar exposure, realized 48h beta to BTC and the point-in-time equal-weight crypto market was moderately negative.

The result was not a one-coin artifact, but dollar-neutral construction did not eliminate realized market exposure.

The negative results define the boundaries of the claim

Observed failures

- Cycle-A medium-horizon momentum failed 2023–2024 validation
- Reversal was statistically strong but economically non-tradable at 20 bps
- Trading-activity conditioning did not produce a stable final refinement
- 3m formation failed sharply in continuation extension
- Holdout economics degraded materially versus internal evidence

What remains uncertain

- Spot data research does not fully model borrow, funding, margin, or capacity
- Constant 20-bps cost is a deliberate simplification
- Final untouched sample is only 19 calendar months
- Overlapping multi-week cohorts reduce independent episode count
- Finalists are highly correlated implementations of one factor

The claim must stay inside these boundaries: continuation-factor generalization with positive cost headroom, not proof of live deployability or a perfect universal market anomaly.

The defensible result is continuation-factor generalization

Generalization

3 / 3 frozen archetypes passed the untouched holdout rule

Economics

all break-even costs remained above the locked 20-bps assumption

Breadth

roughly 29–31 active names and low single-name concentration

Final technical claim: a multi-week cross-sectional continuation architecture, selected using only pre-2025 evidence, generalized into the untouched 2025–July 2026 sample across three frozen implementations with positive execution-cost headroom.

Not claimed: three independent alphas, uniform performance through time, complete market-factor neutrality, or fully modeled live execution